TASK-6: Automated Logging using Triggers & Views

AIM: To develop an automated logging system using MySQL triggers and views that records every INSERT and UPDATE operation and generates daily activity reports.

Algorithm:

1. Create a database and required tables (transactions, audit_log).
2. Create triggers to log INSERT and UPDATE operations automatically.
3. Create a view to summarize daily activity from audit logs.
4. Build a Node.js server to handle insert, update, and fetch requests.
5. Design a simple frontend to send data and display logs/reports.

Code:

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Audit Logging System</title>
</head>
<body>
<h2>Transaction System</h2>
<input id="name" placeholder="User Name">
<input id="amount" placeholder="Amount">
<button onclick="add()">Add</button>
<button onclick="update()">Update ID 1</button>
<h2>Logs</h2>
<button onclick="loadLogs()">Load Logs</button>
<table border="1">
  <thead>
    <tr>
      <th>ID</th>
      <th>Txn ID</th>
      <th>Action</th>
```

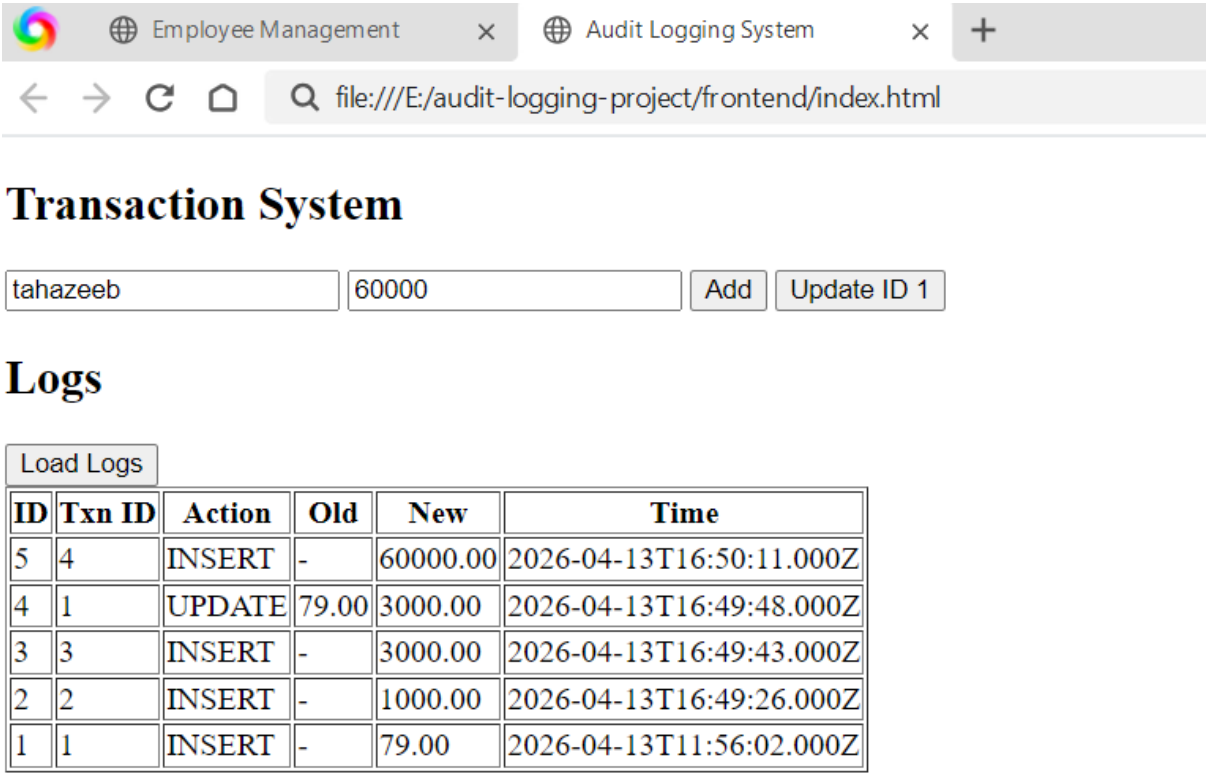
```
            <th>Old</th>
            <th>New</th>
            <th>Time</th>
        </tr>
    </thead>
    <tbody id="logs"></tbody>
</table>
<script>
const API = "http://localhost:3000";
function add() {
    fetch(`${API}/transaction`, {
        method: "POST",
        headers: {"Content-Type": "application/json"},
        body: JSON.stringify({
            user_name: document.getElementById("name").value,
            amount: document.getElementById("amount").value,
            action_type: "PAYMENT"
        })
    }).then(r => r.json()).then(alert);
}
function update() {
    fetch(`${API}/transaction/1`, {
        method: "PUT",
        headers: {"Content-Type": "application/json"},
        body: JSON.stringify({
            amount: document.getElementById("amount").value
        })
    }).then(r => r.json()).then(alert);
}

function loadLogs() {
```

```
  fetch(`${API}/logs`)
  .then(r => r.json())
  .then(data => {
    let rows = "";
    data.forEach(l => {
      rows += `
        <tr>
          <td>${l.log_id}</td>
          <td>${l.transaction_id}</td>
          <td>${l.action}</td>
          <td>${l.old_amount || "-"}</td>
          <td>${l.new_amount || "-"}</td>
          <td>${l.log_time}</td>
        </tr>
      `;
    });
    document.getElementById("logs").innerHTML = rows;
  });
}
</script>

</body>
</html>
```

Output:



ID	Txn ID	Action	Old	New	Time
5	4	INSERT	-	60000.00	2026-04-13T16:50:11.000Z
4	1	UPDATE	79.00	3000.00	2026-04-13T16:49:48.000Z
3	3	INSERT	-	3000.00	2026-04-13T16:49:43.000Z
2	2	INSERT	-	1000.00	2026-04-13T16:49:26.000Z
1	1	INSERT	-	79.00	2026-04-13T11:56:02.000Z

Result: The system successfully logs all INSERT and UPDATE operations automatically using triggers. Audit records store complete details, ensuring proper tracking of data changes. The daily activity view provides a clear summary of operations performed each day**.**